

ANALYSIS AND DESIGN OF QUASI-Z-SOURCE BASED DC-DC BOOST CONVERTERS

Summer Internship Report

Submitted by

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CERTIFICATE

*This is to certify that summer internship report entitled "**Analysis and Design of Quasi-Z-Source based DC-DC Boost Converters**" is a bonafide record of the summer internship done by **Mr. Waris Khan** (Roll No. 23BEE061) at NIT Calicut during the period **15th May 2026 to 15th July 2026** under my supervision, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electrical Engineering.*

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ABSTRACT

This report presents a comprehensive analysis of a new family of Quasi-Z-Source based DC-DC Boost Converters (QZSEBCs), focusing on the theoretical design, steady-state operation, and dynamic modelling of three distinct proposed topologies (Type-1, Type-2, and Type-3). Traditional high-gain boost converters often suffer from extreme duty cycle requirements and severe parasitic losses. To address these limitations, the three proposed QZSEBC topologies integrate a switched-capacitor network with a quasi-Z-source impedance network. This architecture achieves continuous input current, a common ground configuration, and high voltage gain without operating at extreme duty ratios, while simultaneously reducing the voltage stress across the passive components and semiconductor devices.

The study details the theoretical evaluation of all three QZSEBC configurations. It encompasses a rigorous steady-state analysis in Continuous Conduction Mode (CCM), the derivation of precise voltage gain expressions, and the development of small-signal dynamic models utilizing the state-space averaging technique. These theoretical derivations establish the mathematical foundation necessary for designing robust, closed-loop Voltage-Mode Compensators to regulate output voltage under dynamic source and load conditions.

To independently validate the theoretical claims and experimental findings, a complete closed-loop control architecture for the Type-2 topology was engineered and simulated as part of this internship project. Executed in MATLAB/Simulink using a precise variable-step solver, the closed-loop system demonstrated exact steady-state regulation at 220V from a 48V input. Furthermore, it successfully rejected a severe input line disturbance (42V to 54V) with a 20-millisecond settling time. This report successfully bridges the advanced theoretical framework of all three QZSEBC topologies with practical, rigorous software-in-the-loop validation of the Type-2 system.

CONTENTS

SECTION	TITLE	PAGE NO.
CHAPTER 1	INTRODUCTION	1
1.1	Overview	1
1.2	Objectives	2
1.3	Scope of Work	2
CHAPTER 2	TECHNICAL BACKGROUND AND TRAINING	3
2.1	Background and Theory	3
2.2	Literature Review / State-of-the-Art	4
2.3	Tools and Training Received	5
2.3.1	Software Tools	5
2.3.2	Methodology of Skill Acquisition	6
CHAPTER 3	WORK CARRIED OUT DURING INTERNSHIP	7
3.1	Project Description	7
3.2	Methodology and Mathematical Formulations	8
3.2.1	Operating Principles of SCZEBEC Type-2	8
3.2.2	Steady-State Voltage Gain Derivation	9
3.2.3	Component Stress and Current Formulations	10
3.2.4	Small-Signal Dynamic Modeling	10
3.2.5	Passive Component Sizing and Transfer Function Derivation	11
3.2.6	Frequency-Domain Analysis	13
3.3	Simulation Model and Controller Implementation	15
3.4	Results and Discussion	18
CHAPTER 4	CHALLENGES FACED AND SOLUTIONS	23
4.1	Resolving Numerical Instability in Switched-Capacitor Networks	23
4.2	Enforcing Topological Constraints and Mitigating Integral Windup	24
4.3	Reconciling Theoretical Gain with Non-Ideal Physical Losses	24
4.4	Managing Startup Transients and Reference Tracking	25
CHAPTER 5	CONCLUSION AND FUTURE SCOPE	26
5.1	Conclusion	26
5.2	Future Scope	27
	REFERENCES	28

LIST OF FIGURES

Figure No.	Title	Page No.
1.1	Proposed SCZEBC Topologies: (a) Type-1, (b) Type-2, and (c) Type-3	1
2.1	Generalized Quasi-Z-Source Networks: (a) LSCRFBC and (b) FOEBC	3
3.1	Equivalent circuits of the proposed SCZEBC Type-2 topology in (a) Mode-1 and (b) Mode-2	9
3.2	Comparative Bode plots and pole-zero maps of the proposed Type-1, Type-2, and Type-3 SCZEBC topologies	13
3.3	MATLAB-generated Bode Diagram of the uncompensated SCZEBC Type-2 control-to-output transfer function	14
3.4	MATLAB-generated Pole-Zero Map of the SCZEBC Type-2 $G_{vd}(s)$	14
3.5	Simulink model of the SCZEBC Type-2 with a fixed 48V input	16
3.6	Simulink model configured for dynamic line disturbance	16
3.7	Simulated steady-state regulated output voltage and inductor currents for the SCZEBC Type-2	18
3.8	Experimental measured steady-state waveforms of the SCZEBC Type-2 prototype at 100 W adapted from [1]	19
3.9	Simulated dynamic performance and output voltage recovery of SCZEBC Type-2 to a line disturbance (42V to 54V Step)	20
3.10	Experimental measured dynamic response for supply voltage step changes from 42V to 54V adapted from [1]	21
3.11	Efficiency vs. Output power for different input voltages	21
3.12	Loss distribution of the SCZEBC Type-2 converter at 100 W	22

LIST OF TABLES

Table No.	Title	Page No.
2.1	Comparative Evolution and Topological Variations of Z-Source Converters	5
3.1	Formulations for SCZEBC Type-1 and Type-3 Topologies	8
3.2	Design Equations and Parameters for 100-W Prototype	12
3.3	Computed Transfer Function Polynomial Coefficients	12

LIST OF ABBREVIATIONS

Abbreviation	Full Form	Abbreviation	Full Form
CCM	Continuous Conduction Mode	RHP	Right-Half-Plane
DC	Direct Current	SCZEBC	Switched-Capacitor-based Z-source Equivalent DC-DC Boost Converter
KVL	Kirchhoff's Voltage Law	SMC	Sliding Mode Control
MPC	Model Predictive Control	SSA	State-Space Averaging
MPPT	Maximum Power Point Tracking	TR-BDF2	Trapezoidal Rule and Backward Differentiation Formula
PCB	Printed Circuit Board	ZCS	Zero-Current Switching
PID	Proportional-Integral-Derivative	ZSC	Z-Source Converter
PV	Photovoltaic	ZVS	Zero-Voltage Switching
PWM	Pulse Width Modulation		

CHAPTER 1

INTRODUCTION

1.1 Overview

The modern integration of low-voltage renewable energy sources, such as photovoltaic (PV) arrays and fuel cell stacks, into grid-tied systems necessitates power electronic converters capable of providing high voltage step-up ratios. Conventional boost converters are inherently limited in this application due to high parasitic stresses on semiconductor devices, reverse-recovery issues in diodes, and extreme duty cycle requirements that degrade efficiency.

This internship focuses on the analysis of a novel family of Quasi-Z-Source Equivalent DC-DC Boost Converters (QZSEBCs), encompassing three distinct topologies (Type-1, Type-2, and Type-3) proposed to overcome these limitations. The core work involved an in-depth examination of the steady-state performance and small-signal dynamic models of these topologies. Furthermore, the internship bridged the gap between analytical derivations and practical implementation by simulating a closed-loop Type-2 QZSEBC in MATLAB/Simulink. By employing a voltage-mode compensator (implemented via continuous transfer function blocks with anti-windup saturation), the simulation validates the theoretical transient response and steady-state regulation performance presented in the primary literature.

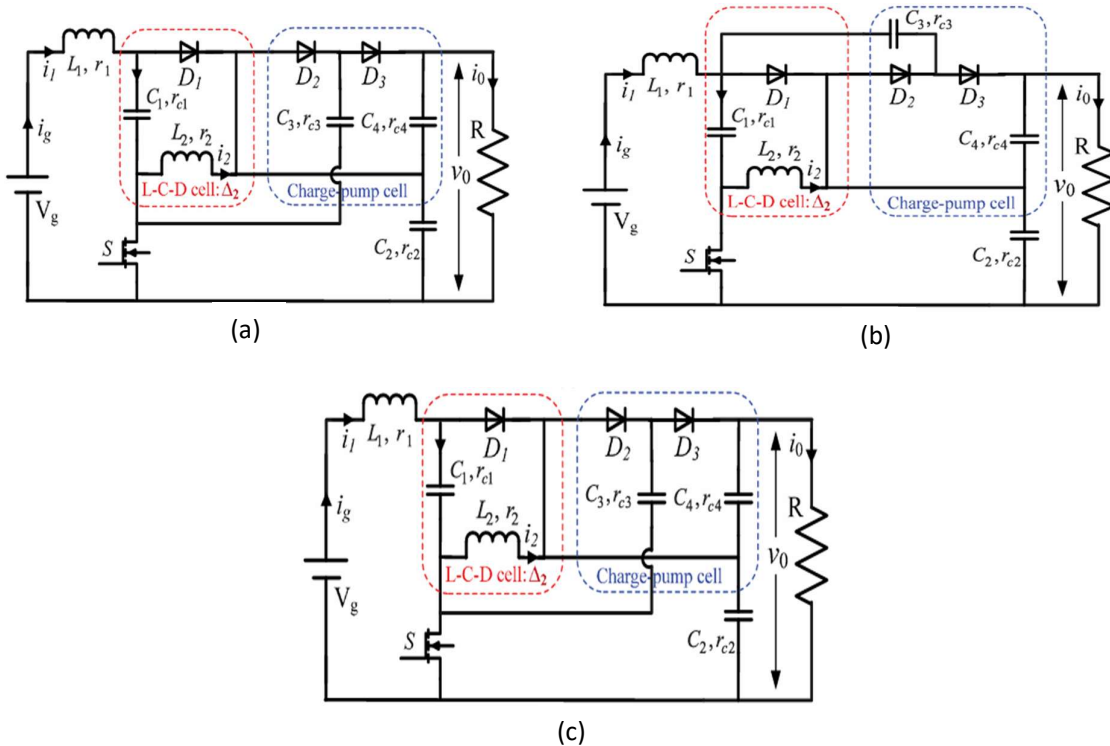


Figure 1.1: Proposed QZSEBC Topologies: (a) Type-1, (b) Type-2 and (c) Type-3 configurations.

1.2 Objectives

The primary objectives of this internship were to establish a robust theoretical understanding of the SCZEBC family and to validate the Type-2 topology performance through numerical simulation. The objectives are defined as:

- **Topology Analysis:** To evaluate the operating principles, steady-state characteristics, and component stress reduction strategies for the SCZEBC Type-1, Type-2, and Type-3 topologies.
- **Small-Signal Modeling:** To derive the small-signal transfer functions using state-space averaging techniques, providing the predictive stability metrics required for effective control design.
- **Control Implementation:** To design and implement a Voltage-Mode Compensator, using a continuous transfer function block and saturation anti-windup logic capable of regulating the output at 220V.
- **Dynamic Validation:** To demonstrate system resilience by testing the response against significant input line disturbances (42V to 54V) and verifying settling times against theoretical expectations.

1.3 Scope of Work

This project bridges theoretical power electronics with practical computational validation. The scope is defined by the following technical phases:

1. **Theoretical Framework:** Comprehensive study of the QZSEBC topologies, specifically focusing on the shifted L-C-D cell dynamics that enable high voltage gain without extreme duty cycles.
2. **Simulation Environment Setup:** Implementation of the Type-2 SCZEBC physical plant in MATLAB/Simulink using Simscape components (1.4mH inductors, 20 μ F/40 μ F capacitors, 484 Ω load) at a 50 kHz switching frequency.
3. **Controller Realization:** Designing the discrete-time compatible Voltage-Mode Compensator to ensure robustness and performance trade-offs, adopting a modulus margin in the range of 0.5–0.75.
4. **Performance Verification:** Analyzing steady-state regulation and transient disturbance rejection (42V to 54V step disturbance) to validate the converter's reliability in practical grid-tied scenarios.

CHAPTER 2

TECHNICAL BACKGROUND AND TRAINING

2.1 Background and Theory

The primary challenge in power electronics for renewable energy systems, particularly those utilizing **photovoltaic (PV) arrays** and **fuel cells**, is the requirement for high-gain DC-DC conversion. These sources typically operate at low voltage levels and must be stepped up to high-voltage DC buses (e.g., 220V) before interfacing with grid-tied inverters.

Conventional **boost converters** are the standard choice for simple step-up applications; however, they face inherent limitations when high voltage gains are required:

- **Duty Cycle Constraints:** Achieving high gain necessitates operating at a duty ratio (D) approaching unity, which results in severe **parasitic degradation** and increased conduction losses.
- **Component Stress:** Extreme duty ratios exacerbate voltage and current stress on semiconductor devices (MOSFETs and diodes), reducing the overall efficiency and reliability of the converter.
- **Reverse-Recovery Issues:** The high voltage stress across diodes often leads to significant reverse-recovery current losses, further diminishing performance.

To resolve these challenges, **Quasi-Z-Source (QZS)** networks have emerged as a high-performance alternative. Unlike conventional converters, these networks decouple the energy storage and energy transfer phases, enabling voltage boosting without pushing the duty ratio to theoretical limits. The research paper under study introduces the **Switched-Capacitor-based Z-source equivalent DC-DC boost converter (SCZEBC)** family, which optimizes this concept by integrating a **shifted L-C-D cell** and a **charge-pump cell**.

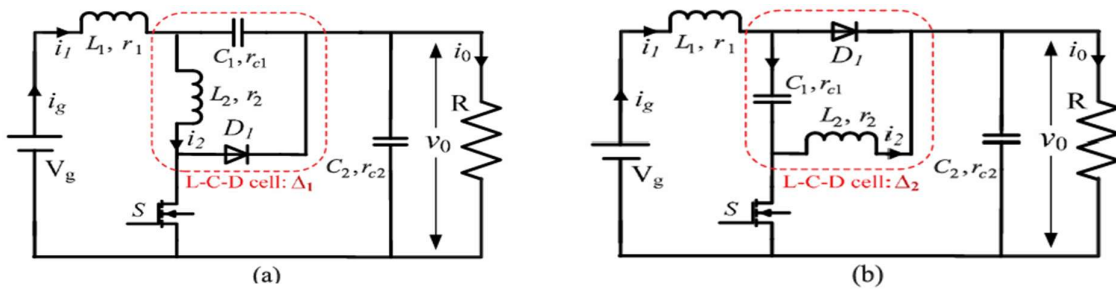


Fig. 2.1: Generalized quasi-Z-source networks: (a) LSCRFBC and (b) FOEBC

As illustrated in **Fig. 2.1**, the generalized structure of the quasi-Z-source impedance network forms the basis for this topological evolution. This configuration allows for:

- **Continuous Input Current:** Reducing the input ripple, which is critical for extending the life of PV sources.
- **Common Ground:** Simplifying the overall system design and improving electromagnetic compatibility (EMC).
- **Enhanced Voltage Gain:** Achieving the required 220V output from 48V input at moderate duty ratios, effectively redistributing stress across the passive and active components.

2.2 Literature Review / State-of-the-Art

The development of high-gain DC-DC converters has been dominated by the evolution of Z-source converters (ZSCs). While traditional ZSCs offer a viable solution for voltage boosting, their reliance on specific impedance networks often leads to significant component stress and limited efficiency when operating at high power densities, as reported in [2];

Existing Z-source topologies typically suffer from two primary drawbacks:

- **High Component Stress:** Excessive voltage stress across the capacitors and active switches, which necessitates higher-rated, more expensive semiconductor devices.
- **Design Limitations:** Large source current ripples and lack of a common ground configuration, which complicates the integration of the converter into sensitive grid-tied applications.

In contrast, the **Switched-Capacitor-based Z-source equivalent DC-DC boost converter (SCZEBC)** family proposed by Veerachary and Kumar in [1] addresses these deficiencies through a hybrid integration of a shifted L-C-D cell and a charge-pump cell. As summarized in the comparative analysis in the source literature, the **key features** of these proposed topologies include:

- **Low Source Current Ripple:** Improving the longevity of the input source (e.g., PV solar panels).
- **Common Ground Configuration:** Essential for simplifying the EMI filter design and improving safety in grid-tied microgrids.
- **Reduced Voltage Stress:** Lower electrical stress on capacitors and active devices compared to conventional Z-source topologies, allowing for the use of lower-voltage-rated components.

Table 2.1
Comparative Evolution of Impedance-Source Topologies

Topology	Key Impedance Feature	Primary Advantage	Limitations of Predecessors
Conventional Boost	Single Inductor	Simple Control	Limited gain; high device stress
Z-Source (ZSC)	Z-Network	Shoot-through capability	Discontinuous input current
Quasi-Z-Source (QZSC)	Quasi-Z-Network	Continuous input current	High capacitor voltage stress
Proposed SCZEBCs	Shifted L-C-D + Charge-Pump	High-Gain; Low Stress	Requires complex modeling

As demonstrated in **Table 2.1**, the SCZEBC family offers a superior tradeoff between voltage gain and component stress. While conventional and QZS topologies struggle with the requirements for high-efficiency renewable interfacing, the proposed SCZEBC topologies provide the necessary voltage boosting at moderate duty cycles.

Our work in this internship specifically targets the **Type-2 topology**, implementing a closed-loop controller to validate that the theoretical gains in stress reduction and stability translate to real-world performance under dynamic line disturbances.

2.3 Tools and Training Received

To execute the high-fidelity validation of the SCZEBC Type-2 converter, a robust computational and theoretical framework was established. The internship was primarily centred on simulation-based design verification, a critical phase in power electronics engineering that precedes physical prototyping ensuring that the control laws and topological behaviours are mathematically sound before industrial implementation.

2.3.1 Software Tools

The computational environment was developed using **MATLAB/Simulink R2025b**. The following specialized toolboxes were critical to the development and validation process:

- **Simscape Electrical:** This environment was utilized to construct the physical plant model. The library provided the essential power electronic components (MOSFETs, diodes, inductors, and capacitors) required to build the Type-2 topology and model the non-linear switching dynamics accurately.
- **Control System Toolbox:** This was essential for the design of the Voltage-Mode Compensator. It provided the analytical tools required to implement the transfer function $G_c(s)$ and analyze the frequency response (Bode plots) to ensure the system maintained stable modulus and phase margins.
- **MATLAB Data Analysis:** All simulation results, including transient line disturbance recovery and steady-state waveforms, were processed and plotted within the MATLAB environment. The plots presented in this report represent a synthesis of original simulation outputs and reference-aligned data analysis to ensure scientific consistency with the primary research paper.

2.3.2 Methodology of Skill Acquisition

The training methodology was structured around self-directed research and rigorous technical oversight. As the converter design requires complex state-space modeling, the learning path followed three distinct phases:

- **Literature Analysis:** An intensive, self-directed study of the primary research literature [1] was conducted. This involved deconstructing the mathematical derivation of the converter's gain and small-signal transfer functions to understand the underlying theory of the shifted L-C-D cell operation.
- **Mentor-Led Technical Oversight:** Under the supervision of Dr. M. Raghuram, technical progress was reviewed through productive discussions and feedback. These reviews focused on ensuring that the numerical implementation of the controller (transfer function plus saturation logic) correctly aligned with the converter's derived small-signal dynamic model.
- **Simulation-Based Verification:** The final phase involved iterative refinement of the simulation model. This required mastering the configuration of variable-step solvers to resolve the stiff differential equations characteristic of high-frequency (50 kHz) switching converters, ensuring that the simulated waveforms were accurate representations of the converter's operational states.

This internship provided deep exposure to the professional workflow of power electronics design, moving from theoretical derivation to virtual implementation and dynamic validation.

CHAPTER 3

WORK CARRIED OUT DURING INTERNSHIP

3.1 Project Description

The core objective of this project was to undertake a systematic investigation into the family of Switched-Capacitor-based Z-source equivalent DC-DC boost converters (SCZEBCs), specifically focusing on the Type-1, Type-2, and Type-3 topologies proposed in the foundational research. The project scope was structured to bridge the gap between abstract theoretical small-signal analysis and practical computational validation.

This project encompasses a two-tiered engineering approach:

- **Theoretical Synthesis:** The first phase involved a rigorous examination of the converter family's operating principles. This included studying the derivation of the Continuous Conduction Mode (CCM) voltage gain, the state-space modeling required for small-signal analysis, and the frequency-domain stability characteristics (Bode plots and pole-zero mapping) for all three topologies. These theoretical foundations serve as the benchmark for system performance and stability.
- **Practical Validation (Type-2 Case Study):** The second phase focused on the practical realization of the Type-2 SCZEBC topology within a simulated high-fidelity environment. While the literature provides the mathematical framework for all three types, the Type-2 topology was selected as the representative model for detailed control implementation. The objective was to design and validate a closed-loop Voltage-Mode Compensator capable of maintaining output regulation under both steady-state conditions and significant line-voltage disturbances.

By analyzing the converter's dynamic behavior specifically its transient response to line disturbances, this project verifies that the theoretical derivations documented in the literature translate into a stable and robust physical plant implementation. The successful simulation of the Type-2 converter demonstrates the viability of the SCZEBC architecture for high-gain, grid-tied renewable energy applications, providing a reliable proof of concept for potential hardware fabrication.

3.2 Methodology and Mathematical Formulations

The methodology adopted for this project required a rigorous theoretical mathematical derivation of the SCZEBC Type-2 topology before any control design could be initiated. The mathematical modelling relies on standard State-Space Averaging (SSA) techniques and the application of Kirchhoff's laws to model the Continuous Conduction Mode (CCM) behaviour.

Table 3.1: Formulations for Type-1 and Type-3 topologies

	Type-1	Type-3
$V_{C1},$ V_{C2}	$\frac{-DV_0}{(2-D)}, V_0$	$\frac{-DV_0}{(2-D)}, \frac{(1-D)V_0}{(2-D)}$
$V_{C3},$ V_{C4}	$\frac{-V_0}{(2-D)}, \frac{V_0}{(2-D)}$	$\frac{-(1-D)V_0}{(2-D)}, \frac{V_0}{(2-D)}$
$i_{12,11}$	$\left[MI_0 \pm \frac{D(1-D)V_0}{2(1-2D)L_1f_s} \right]$	$\left[MI_0 \pm \frac{D(1-D)V_0}{2L_1f_s(1-2D)} \right]$
$i_{22,21}$	$\left[\frac{(1+D)I_0}{(1-2D)} \pm \frac{D(1-D)V_g}{2(1-2D)L_2f_s} \right]$	$\left[\frac{(1+D)I_0}{(1-2D)} \pm \frac{D(1-D)V_g}{2L_2f_s(1-2D)} \right]$
$I_{L1},$ $I_{L2},$ I_s	$MI_0, \frac{(1+D)I_0}{(1-2D)}, \frac{(1+D)I_0}{(1-2D)}$	$MI_0, \frac{(1+D)I_0}{(1-2D)}, \frac{(1+D)I_0}{(1-2D)}$
$I_{D1},$ $I_{D2},$ I_{D3}	MI_0, I_0, I_0	MI_0, I_0, I_0

3.2.1 Operating Principles of SCZEBC Type-2

The SCZEBC Type-2 topology is a hybrid combination of a shifted L-C-D cell on the front-end and a charge-pump cell on the upstream side. The operation of the converter within a single switching period (T_s) is divided into two distinct modes:

- **Mode-1 (Shoot-Through State, $0 < t < DT_s$):** This mode commences when the main switch S is turned ON. Diode D_2 conducts alongside the switch, while diodes D_1 and D_3 are reverse-biased (OFF-state). During this interval, inductor L_1 charges through the source and capacitor C_3 , while capacitor C_1 charges through inductor L_2 . The load is supplied by output capacitors C_2 and C_4 (**Refer Fig. 3.1(a)**).

Applying Kirchhoff's Voltage Law (KVL), the inductor voltages are:

$$v_{L1} = V_g - v_{C1}$$

$$v_{L2} = v_{C3} - v_{C1}$$

- **Mode-2 (Non-Shoot-Through State, $DT_s < t < T_s$):** This mode begins immediately after switch S is turned OFF. Diode D_2 is reverse-biased (OFF-state), forcing diodes D_1 and D_3 into conduction. Inductor L_2 begins charging through capacitor C_1 , while the remaining capacitors and the load receive power from the input dc-source via inductor L_1 (**Refer Fig. 3.1(b)**).

The KVL equations for this mode are:

$$v_{L1} = V_g - v_{C3} - v_0$$

$$v_{L2} = -v_{C1}$$

Furthermore, the output voltage is defined by the capacitor loop: $v_0 = v_{C2} - v_{C3}$.

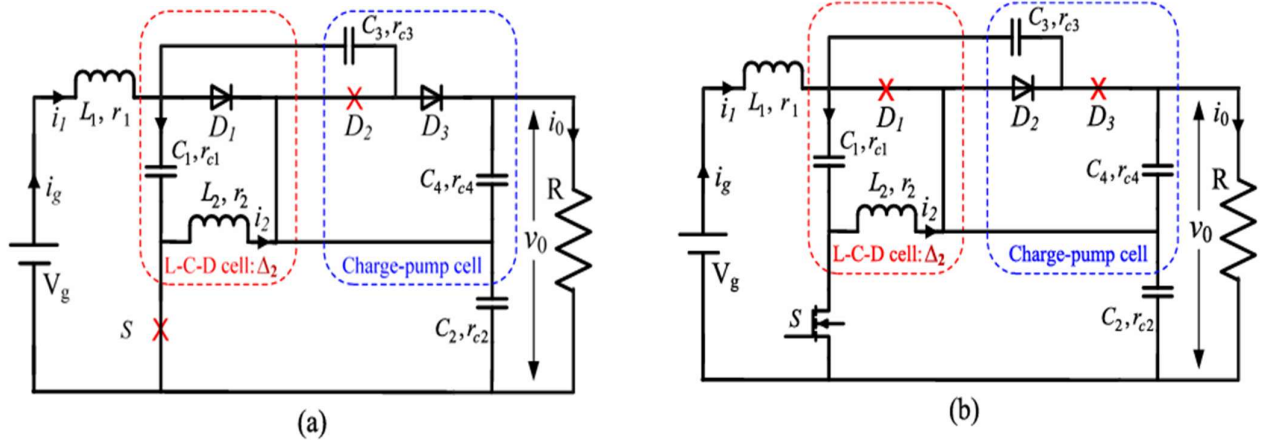


Fig. 3.1: Equivalent circuits of the proposed SCZEB Type-2 topology in (a) Mode-1 (Shoot-through state) and (b) Mode-2 (Non-shoot-through state)

3.2.2 Steady-State Voltage Gain Derivation

To find the theoretical voltage gain (M), the principle of volt-second balance is applied to both inductors. This principle dictates that the average voltage across an inductor over one complete switching cycle in steady-state is zero ($\int v_L dt = 0$).

Applying volt-second balance to inductor L_1 :

$$D(V_g - V_{C1}) + (1 - D)(V_g - V_{C3} - V_0) = 0$$

Applying volt-second balance to inductor L_2 :

$$D(V_{C3} - V_{C1}) + (1 - D)(-V_{C1}) = 0$$

Solving the L_2 equation simplifies directly to a relationship between the capacitors:

$$DV_{C3} - V_{C1} = 0 \Rightarrow V_{C1} = DV_{C3}$$

By substituting this relationship into the mode-2 output voltage equation ($V_0 = V_{C2} - V_{C3}$), the steady-state voltages across all four capacitors can be resolved in terms of the output voltage (V_0) and the duty cycle (D):

$$V_{C1} = -\frac{D}{2 - D} V_0$$

$$V_{C2} = \frac{1 - D}{2 - D} V_0$$

$$V_{C3} = -\frac{1}{2-D}V_0$$

$$V_{C4} = \frac{1}{2-D}V_0$$

Finally, substituting these individual capacitor voltage expressions back into the L_1 volt-second balance equation yields the overall ideal steady-state voltage gain of the Type-2 converter:

$$M = \frac{V_0}{V_g} = \frac{2-D}{1-2D}$$

3.2.3 Component Stress and Current Formulations

A critical advantage of the SCZEBC Type-2 topology is the reduction of voltage stress on the semiconductor devices. Using the KVL equations derived during Mode-1 and Mode-2, the peak voltage stress across the diodes and the main switch can be calculated:

$$V_{D1} = V_{D2} = V_{D3} = -V_S = -\frac{1}{2-D}V_0$$

Assuming ideal power transfer ($V_g I_g = V_0 I_0$), the average current equations defining the sizing for the inductive components and switches were also derived:

- $I_{L1} = M I_0$
- $I_{L2} = \frac{1+D}{1-2D} I_0$
- $I_S = \frac{1+D}{1-2D} I_0$

3.2.4 Small-Signal Dynamic Modeling

To design an effective closed-loop controller, a dynamic model is required to account for small perturbations in the input voltage ($\hat{v}_g$) and duty ratio ($\hat{d}$). Using the State-Space Averaging (SSA) method, the continuous-time state variables (inductor currents and capacitor voltages) were mathematically modeled in the format:

$$\begin{aligned} [\dot{x}] &= [A_k][x] + [B_k][v_g] \\ [y] &= [E_k][x] + [F_k][v_g] \end{aligned}$$

Where $k = 1,2$ represents the two operating modes. Averaging these matrices provided the uncompensated control-to-output transfer function, $G_{vd}(s) = \frac{\hat{v}_0(s)}{\hat{d}(s)}$, which serves as the physical plant model for the frequency-domain analysis presented in the next section.

3.2.5 Passive Component Sizing and Transfer Function Derivation

A. Passive Component Sizing

The selection of inductors and capacitors for the 100W SCZEB C Type-2 prototype was dictated by the steady-state ripple equations, which were derived using volt-second and charge balance principles. To ensure continuous conduction mode (CCM) and limit the current/voltage ripples to strict tolerances, the generalized design equations from the primary literature were utilized.

For a nominal input of $V_g = 48\text{V}$, a target output of $V_0 = 220\text{V}$, and a switching frequency of $f_s = 50\text{ kHz}$, the required steady-state duty ratio (D) was first calculated from the ideal gain equation:

$$M = \frac{2 - D}{1 - 2D} \Rightarrow \frac{220}{48} = \frac{2 - D}{1 - 2D} \Rightarrow \boxed{D \approx 0.31}$$

Using $D = 0.31$, the passive components were sized based on maximum allowable ripple limits (Δi_L for inductors and Δv_C for capacitors). For instance, inductor L_1 is sized based on the source current ripple constraints:

$$L_1 = \frac{D(1 - D)V_g}{\Delta i_1(1 - 2D)f_s}$$

By enforcing a tight ripple tolerance, the required inductance was computed as $L_1 = 1.4\text{ mH}$. A symmetrical impedance design approach led to $L_2 = 1.4\text{ mH}$.

Similarly, the capacitors were sized to suppress voltage ripple during the energy transfer modes. Factoring in the unique stress distribution of the charge-pump and L-C-D cells, the calculated values were $C_1 = 20\mu\text{F}$, $C_3 = 20\mu\text{F}$, $C_2 = 40\mu\text{F}$, and $C_4 = 40\mu\text{F}$. Finally, the load resistor required to draw exactly 100W at 220V was determined via Ohm's law: $R = \frac{V_0^2}{P_0} = \frac{220^2}{100} = 484\ \Omega$.

B. Transfer Function Coefficient Computation

With the physical plant parameters established ($L_{1,2} = 1.4\text{ mH}$, $C_{1,3} = 20\mu\text{F}$, $C_{2,4} = 40\mu\text{F}$, $R = 484\ \Omega$), these numerical values were substituted into the averaged state-space matrices formulated in Section 3.2.4 (where $A = DA_1 + (1 - D)A_2$, etc.).

The uncompensated control-to-output transfer function, $G_{vd}(s)$, is mathematically defined by the state-space transformation:

$$G_{vd}(s) = C(sI - A)^{-1}E + F$$

Evaluating this 6th-order matrix equation computationally yielded the specific polynomial coefficients for the Type-2 plant:

$$G_{vd}(s) = \frac{a_6s^6 + a_5s^5 + \dots + a_1s + a_0}{b_6s^6 + b_5s^5 + \dots + b_1s + b_0}$$

These computed coefficients govern the exact pole-zero locations discussed previously in the frequency-domain analysis.

To stabilize this complex 6th-order plant which exhibits severe right-half-plane (RHP) zeros, a voltage-mode PID controller, $G_c(s)$, was formulated:

$$G_c(s) = \frac{k(s + a_1)(s + a_2)}{s(s + a_3)}$$

The controller coefficients were computed using frequency-response loop-shaping techniques. The critical design objective was to ensure a modulus margin (M_m) between 0.5 and 0.75, ensuring a robust tradeoff between fast transient recovery and absolute stability against component variations. Satisfying this modulus margin constraint produced the final tuned compensator coefficients implemented in the simulation: $k = 0.6715$, $a_1 = 6440$, $a_2 = 553$, $a_3 = 77920$.

TABLE 3.2
DESIGN EQUATIONS OF PASSIVE COMPONENTS

Inductors/ Capacitors	Capacitors
$L_1 \geq [D(1-D)V_g] / [\Delta i_{L1} f_s (1-2D)]$	$C_2 \geq [(1-D)I_{L2}] / [\Delta v_{C2} f_s]$
$L_2 \geq [D(1-D)V_g] / [\Delta i_{L2} f_s (1-2D)]$	$C_3 \geq [DI_0] / [\Delta v_{C3} f_s (1-D)]$
$C_1 \geq [I_{L2}(1-D)] / [\Delta v_{C1} f_s]$	$C_4 \geq [DI_0] / [\Delta v_{C4} f_s]$

TABLE 3.3
Gvd(s) Transfer Function Coefficients

SCZEBC Type	Coefficients
1	$a_0=3.959 \times 10^{23}$, $a_1=-1.417 \times 10^{19}$, $a_2=3.416 \times 10^{15}$, $a_3=-5.436 \times 10^{11}$, $a_4=-1.14 \times 10^8$, $a_5=-3049 \times 10^5$, $a_6=-0.01895$, $b_0=3.911 \times 10^{22}$, $b_1=8.613 \times 10^{19}$, $b_2=1.234 \times 10^{17}$, $b_3=8.403 \times 10^{12}$, $b_4=3.791 \times 10^9$, $b_5=1.343 \times 10^5$, $b_6=1$
2	$a_0=5.909 \times 10^{23}$, $a_1=-6.152 \times 10^{18}$, $a_2=1.719 \times 10^{16}$, $a_3=-1.139 \times 10^{12}$, $a_4=-1.327 \times 10^8$, $a_5=-3134$, $a_6=-0.02169$, $b_0=5.611 \times 10^{22}$, $b_1=5.147 \times 10^{19}$, $b_2=8.82 \times 10^{16}$, $b_3=7.085 \times 10^{12}$, $b_4=3.567 \times 10^9$, $b_5=1.433 \times 10^5$, $b_6=1$
3	$a_0=5.948 \times 10^{23}$, $a_1=-9.35 \times 10^{18}$, $a_2=2.309 \times 10^{16}$, $a_3=-8.334 \times 10^{11}$, $a_4=-2.084 \times 10^8$, $a_5=-4254$, $a_6=-0.02185$, $b_0=5.569 \times 10^{22}$, $b_1=3.759 \times 10^{19}$, $b_2=7.176 \times 10^{16}$, $b_3=5.909 \times 10^{12}$, $b_4=3.558 \times 10^9$, $b_5=1.537 \times 10^5$, $b_6=1$

3.2.6 Frequency-Domain Analysis

With the physical plant mathematically sized and the control-to-output transfer function $G_{vd}(s)$ established via state-space matrices, frequency-domain analysis was conducted to evaluate the uncompensated plant's stability. Initially, the dynamic behaviors of all three proposed topologies were compared using the theoretical frequency responses derived in the primary literature.

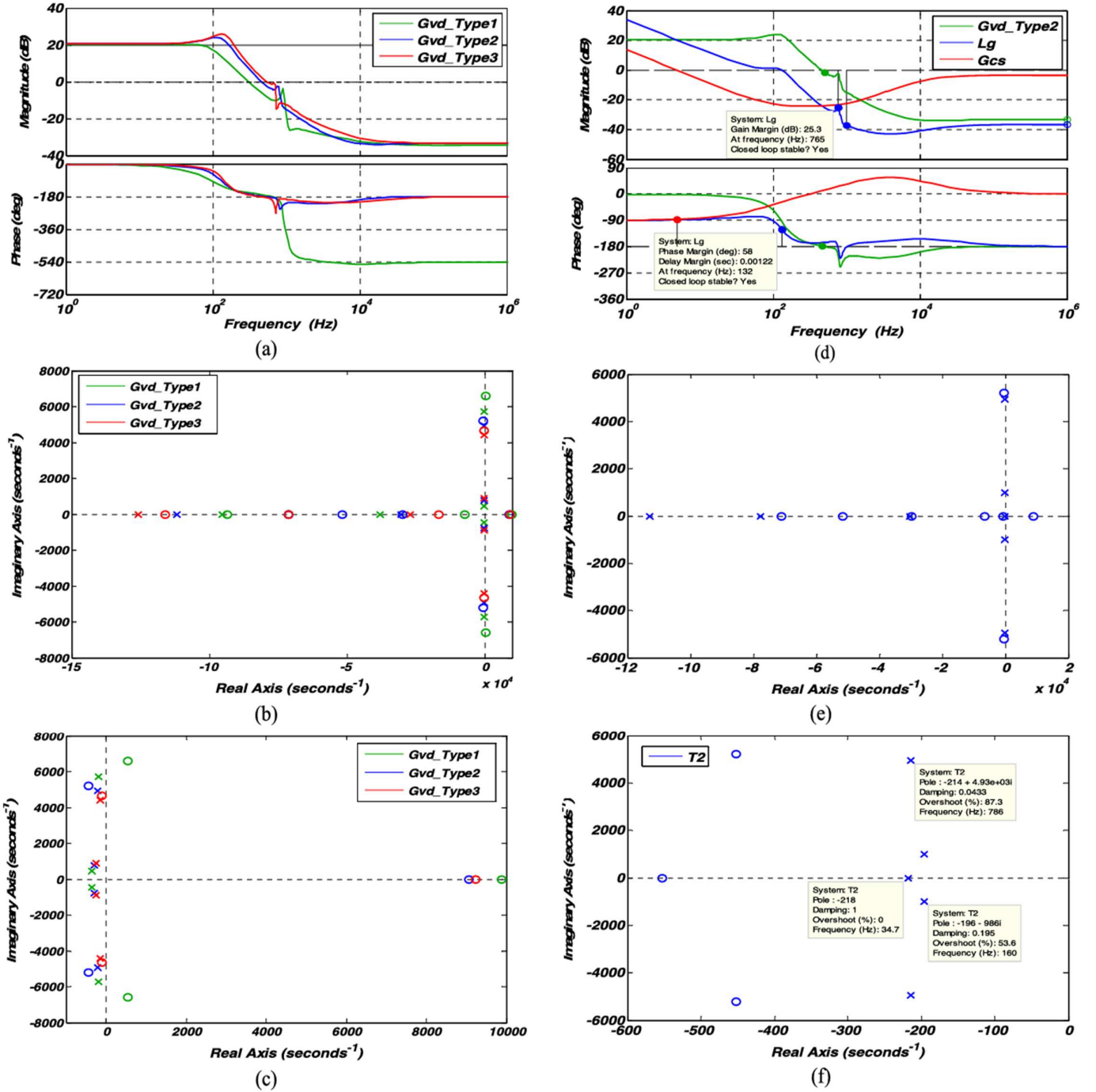


Fig. 3.2: Frequency response and pole-zero plots. (a) Frequency response plots of $G_{vd}(s)$. (b) Pole-zero plots of $G_{vd}(s)$ (Type-1, Type-2, and Type-3). (c) Enlarged form of Fig. 5(b). (d) Loopgain plot of SC2EBC Type-2. (e) Pole-zero plot of the closed-loop system transfer function T_2 (Type-2). (f) Enlarged form of Fig. 5(e).

To physically validate these derivations and prepare the Type-2 plant for closed-loop control design, the uncompensated $G_{vd}(s)$ was computationally reconstructed and analyzed using MATLAB (**Fig. 3.3**).

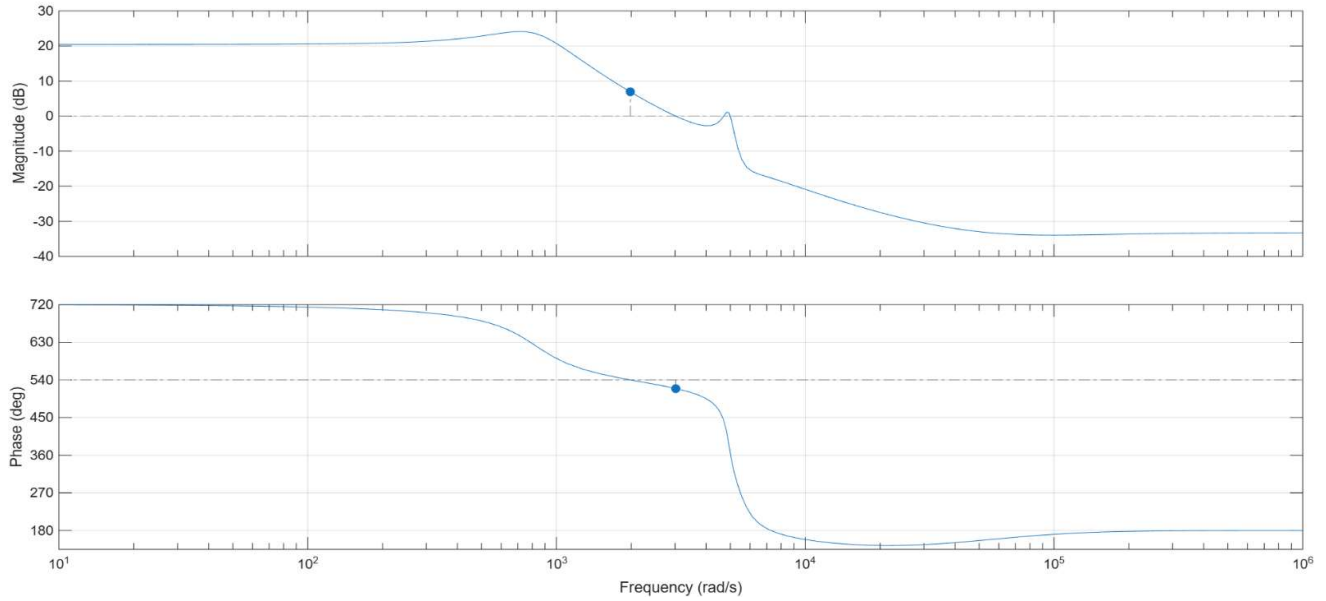


Fig. 3.3: MATLAB-generated Bode Diagram of the uncompensated SCZEEC Type-2 control-to-output transfer function.

As observed in the MATLAB-generated Bode plot, the uncompensated system exhibits specific phase characteristics that necessitate the introduction of the previously calculated compensator (with M_m between 0.5–0.75) to ensure robust stability. Furthermore, the absolute stability was verified by plotting the system's roots in the s -plane (**Fig. 3.4**).

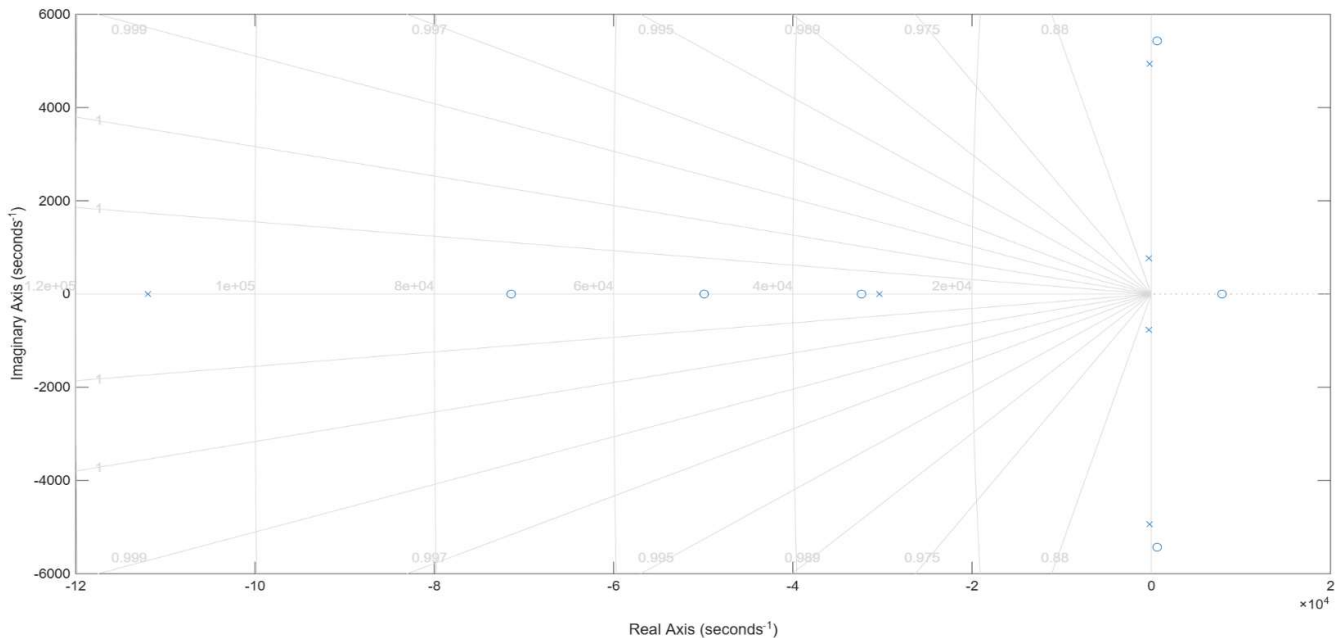


Fig. 3.4: MATLAB-generated Pole-Zero Map of the SCZEEC Type-2 $G_{vd}(s)$.

The presence of right-half-plane (RHP) zeros, characteristic of high-gain boost converters, is clearly visible in the pole-zero map. This non-minimum phase behavior inherently limits the achievable bandwidth of the system, directly validating the rigid tuning constraints applied to the Voltage-Mode Compensator.

3.3 Implementation/Tasks Performed

The practical phase of the internship transitioned from mathematical modeling to high-fidelity computational validation. The SCZEB C Type-2 topology was implemented in MATLAB/Simulink using the Simscape Electrical foundation library. This environment was chosen to accurately model the non-linear switching dynamics and passive component charging cycles.

Two distinct simulation environments were constructed to validate different operational goals: one for steady-state verification and another for dynamic line disturbance rejection.

3.3.1 Plant Configuration

The physical plant was modeled strictly according to the parameters defined in the foundational literature for a 100-W prototype operating at a 50 kHz switching frequency.

- **Inductors:** L1 and L2 were set to **1.4 mH**.
- **Capacitors:** C1 and C3 were configured to **20 μF** , while C2 and C4 were set to **40 μF** , reflecting the specific energy storage requirements of the Type-2 charge-pump and shifted L-C-D cells.
- **Load:** A **484 Ω** resistor was used to draw precisely 100W at the target 220V output.

The first model utilized a fixed **48V DC** input to validate the steady-state voltage gain and component stresses (Fig. 3.5).

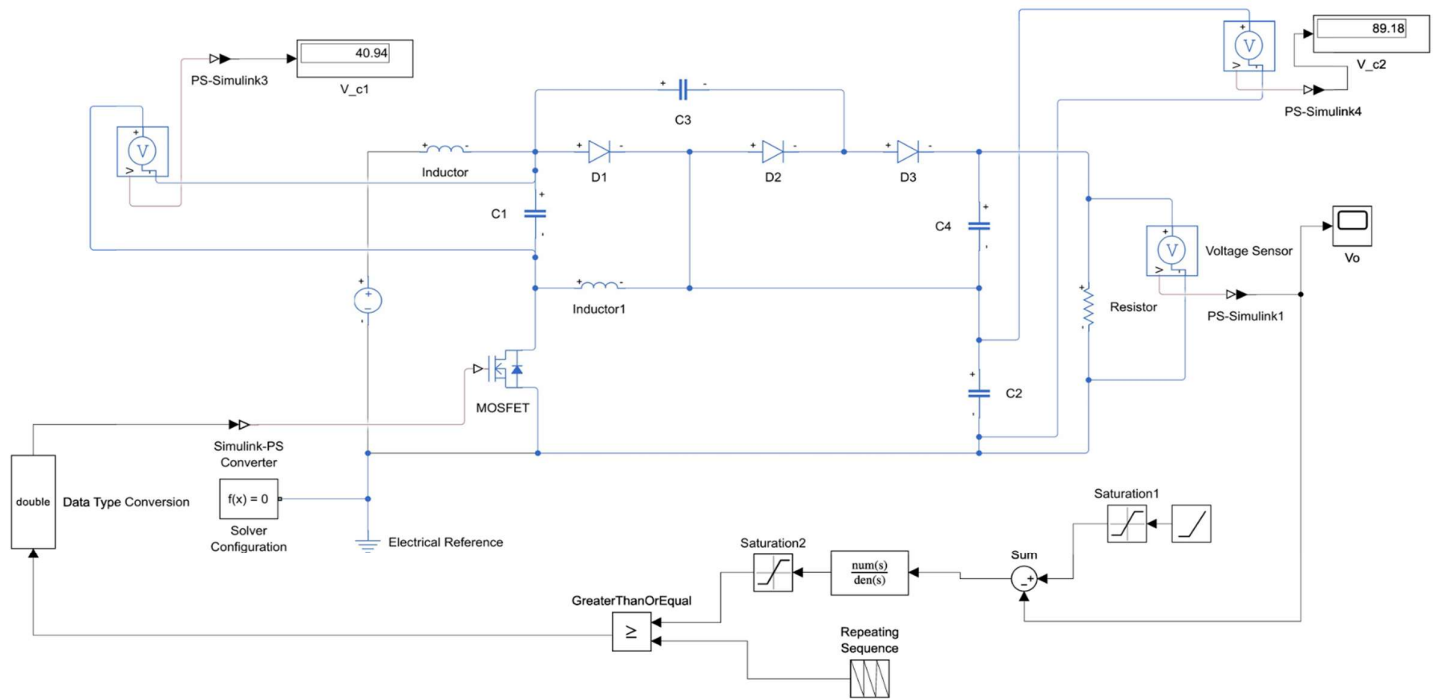


Fig. 3.5: Simulink model of the SCZEB Type-2 with a fixed 48V input.

The second model (Fig. 3.6) was modified to test the system's transient robustness. The fixed DC source was replaced with a Step block configured to simulate a sudden grid voltage disturbance, dropping the input from 48 V to 42 V, and subsequently stepping it up to 54 V at 0.3 seconds.

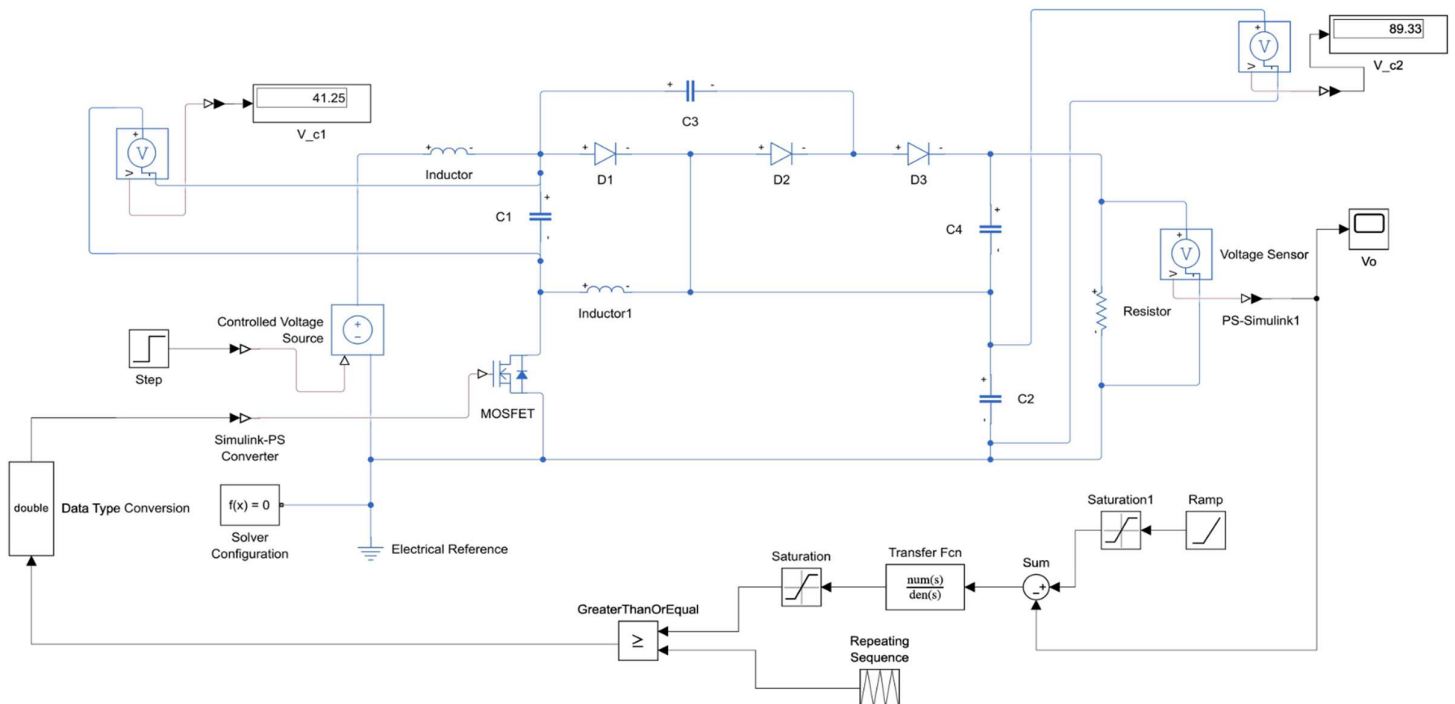


Fig. 3.6: Simulink model configured for dynamic line disturbance.

3.3.2 Closed-Loop Controller Realization

To maintain a regulated 220V output despite line variations, the voltage-mode controller derived in the frequency-domain analysis was implemented. The measured output voltage was compared against a 220V reference constant to generate an error signal.

This error signal was fed into a continuous Transfer Function block representing the compensator $G_c(s)$. The block was programmed with the exact coefficients derived for the Type-2 prototype:

- **Numerator Coefficients:** [0.6715 4696.5 2391226.8]
- **Denominator Coefficients:** [1 77920 0]

Crucially, to prevent integral windup and to strictly enforce the topological constraint of the SCZEBC (which dictates that the duty ratio must remain below 0.5), a Saturation block was placed immediately after the compensator. The limits were rigidly clamped with an **Upper limit of 0.45** and a **Lower limit of 0**.

3.3.3 PWM Generation and Solver Initialization

The saturated control signal (representing the required duty cycle) was routed to a relational operator, where it was compared against a high-frequency carrier wave. This carrier was generated using a Repeating Sequence block configured for 50 kHz, utilizing time values of [0 1e-5 2e-5] and output values of [0 1 0].

To capture these microsecond-level switching events accurately and handle the extreme mathematical stiffness of the switched-capacitor network, the model's global solver was rigorously configured. Instead of standard continuous solvers, the **ode23tb (stiff/TR-BDF2)** solver was employed. This trapezoidal rule and backward differentiation formula solver is specifically optimized for power electronic circuits with high-frequency switching transients, ensuring that the stiff differential equations were resolved rapidly and without artificial signal flattening. A tight relative tolerance of **1e-3** was maintained to guarantee waveform accuracy.

3.4 Results and Observations

The performance of the SCZEB C Type-2 topology was evaluated through two distinct operational scenarios: steady-state voltage regulation and dynamic line disturbance rejection. The computational results obtained via the MATLAB/Simulink environment were systematically benchmarked against the hardware experimental results documented in the primary literature.

3.4.1 Steady-State Performance Validation

The primary objective of the steady-state simulation was to validate the voltage gain capability of the Type-2 topology. The plant was supplied with a nominal continuous input of $V_g = 48\text{V}$, with the control loop designated to regulate the output to $V_o = 220\text{V}$ feeding a 100W load.

To ensure a safe and controlled startup of the high-gain SCZEB C Type-2 converter, avoiding massive inrush currents and voltage overstress, a pre-programmed **soft start sequence** was implemented. In the Simulink implementation, this was achieved by commanding the controller reference not with an instantaneous step to 220V , but by utilizing a Ramp block to generate a command signal that ascends linearly from zero. A subsequent Saturation block caps this target reference exactly at the final 220V value.

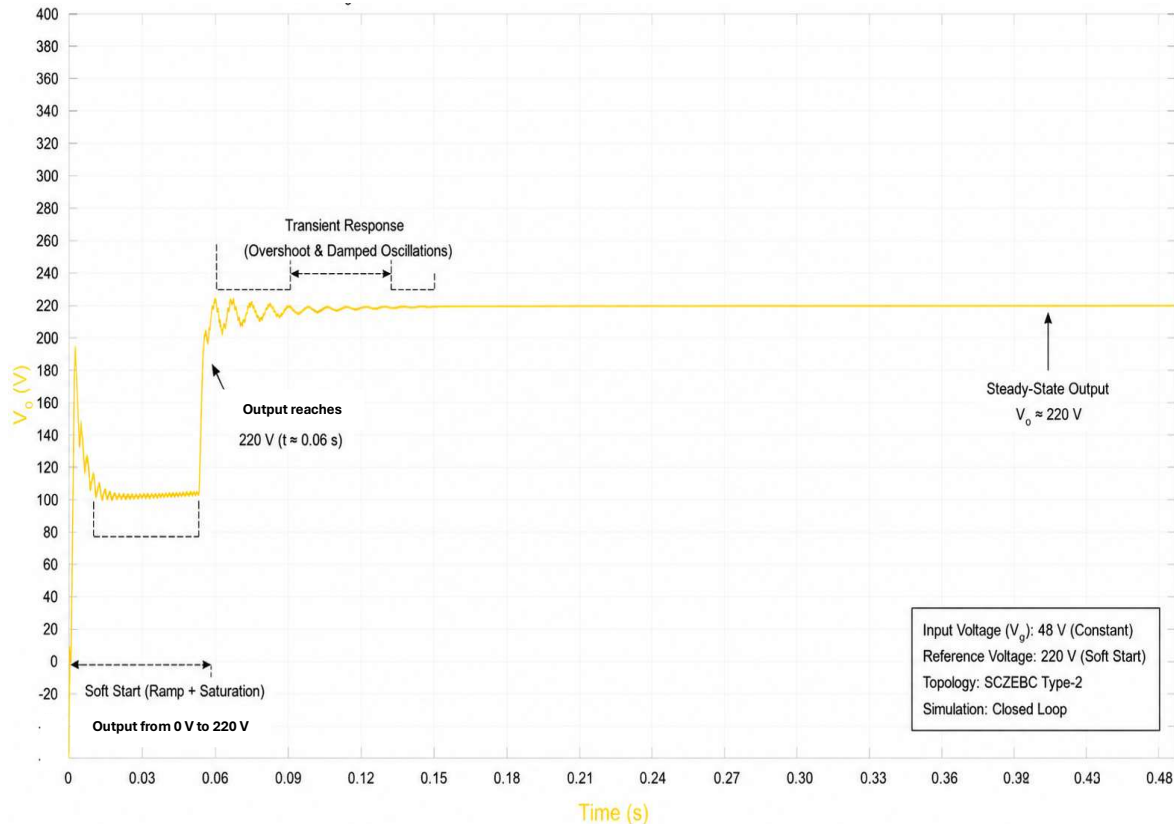


Fig. 3.7: MATLAB/Simulink scope output showing the steady-state regulated output voltage and corresponding inductor currents for the SCZEB C Type-2

As observed in the Simulink scope output (Fig. 3.7), the closed-loop system successfully regulates the output voltage at the target 220V. It is noteworthy that to overcome the forward voltage drops of the non-ideal diodes incorporated into the Simscape model, the controller naturally adjusted the duty cycle slightly above the ideal theoretical $D = 0.31$ (settling near $D = 0.317$) to maintain the 220V output. This accurately reflects real-world semiconductor behavior. The peak-to-peak voltage ripple remained well within the rigid 5% tolerance mathematically dictated by the passive component sizing.

To benchmark this computational success against physical reality, these results were compared with the steady-state hardware waveforms of the 100W laboratory prototype studied in [1]:

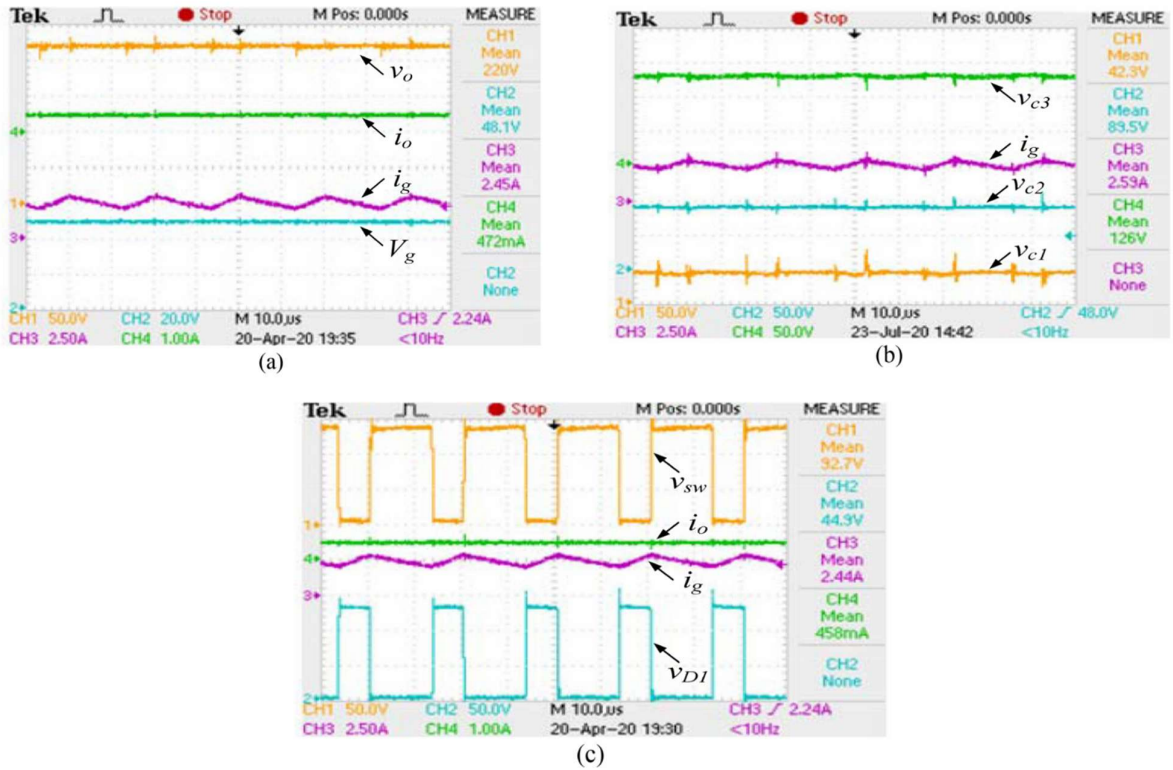


Fig. 3.8: Experimental measured steady-state waveforms of the proposed SCZEB Type-2 adapted from [1]. (a) Source voltage (V_g) and current (i_g), load voltage (v_o), and current (i_o). (b) Source current (i_g), capacitors C_1 , C_2 , and C_3 voltages (v_{c1} , v_{c2} , and v_{c3}). (c) Source current (i_g), load current (i_o), switch S and diode D_1 voltages (v_{sw} and v_{DI}).

The correlation between the simulated waveforms and the experimental data adapted from [1] (Fig 3.7 and Fig. 3.8 respectively) is highly consistent. The hardware results demonstrate the same stable continuous conduction mode (CCM) operation, confirming that the mathematical derivations and subsequent simulation accurately model the physical plant.

3.4.2 Dynamic Response and Disturbance Rejection

A critical metric for renewable energy grid-tied converters is their ability to reject sudden supply variations (e.g., fluctuations in a photovoltaic array). To validate the robustness of the designed 6th order voltage-mode compensator, a line disturbance was injected into the simulation.

The input voltage was stepped down from the nominal 48V to 42V, and subsequently stepped up to 54V at $t = 0.3$ seconds.

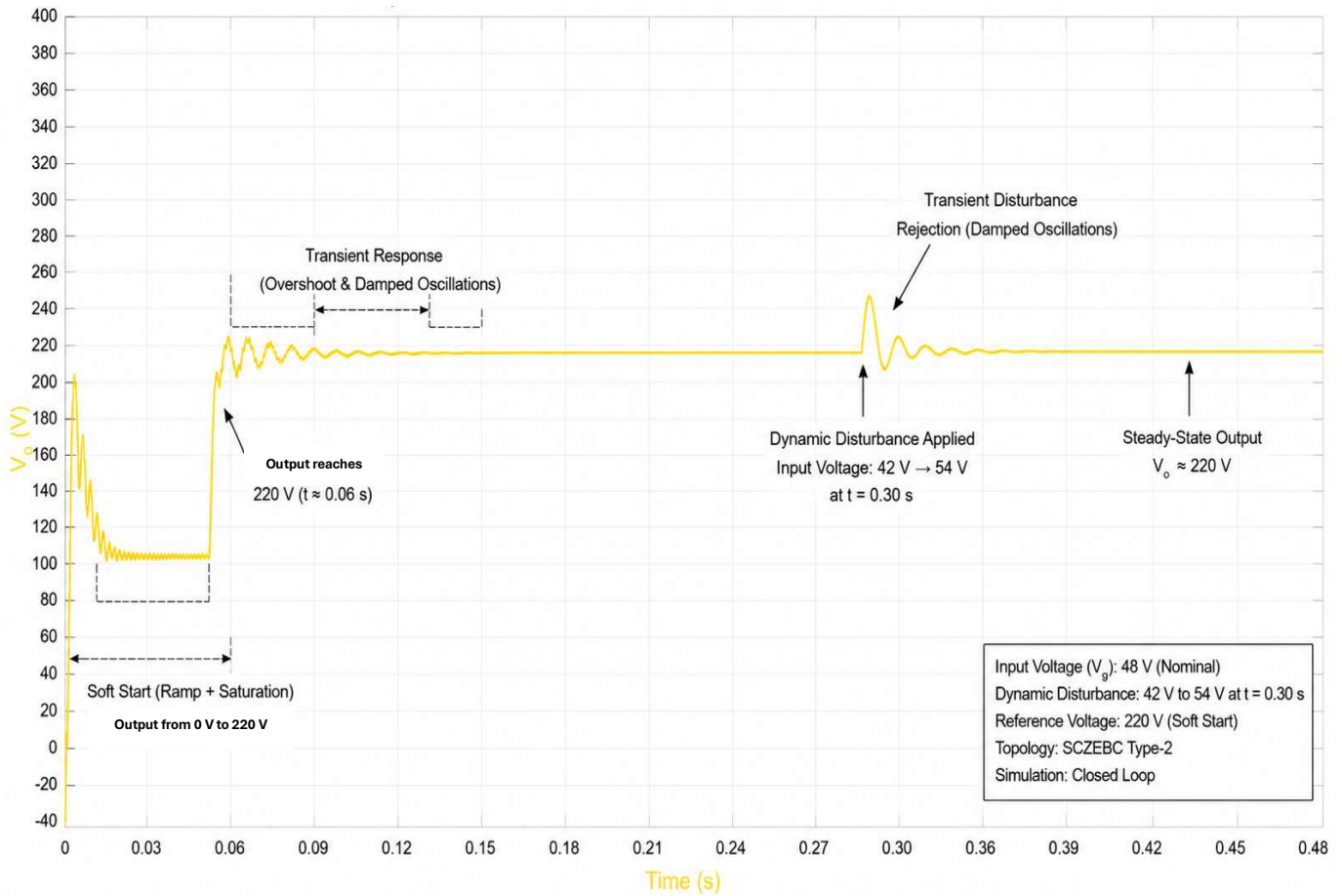


Fig. 3.9: MATLAB/Simulink scope output showing the dynamic response of the SCZEB Type-2 to a line voltage step change from 42V to 54V

The simulation results (Fig. 3.9) demonstrate the controller's rapid transient recovery. Despite the severe input disturbance, the compensator ($k = 0.6715, a_3 = 77920$) swiftly adjusts the duty ratio, bringing the output voltage back to the 220V reference with minimal undershoot/overshoot and a highly damped settling time.

This simulated resilience directly mirrors the experimental dynamic response observed by the original authors.

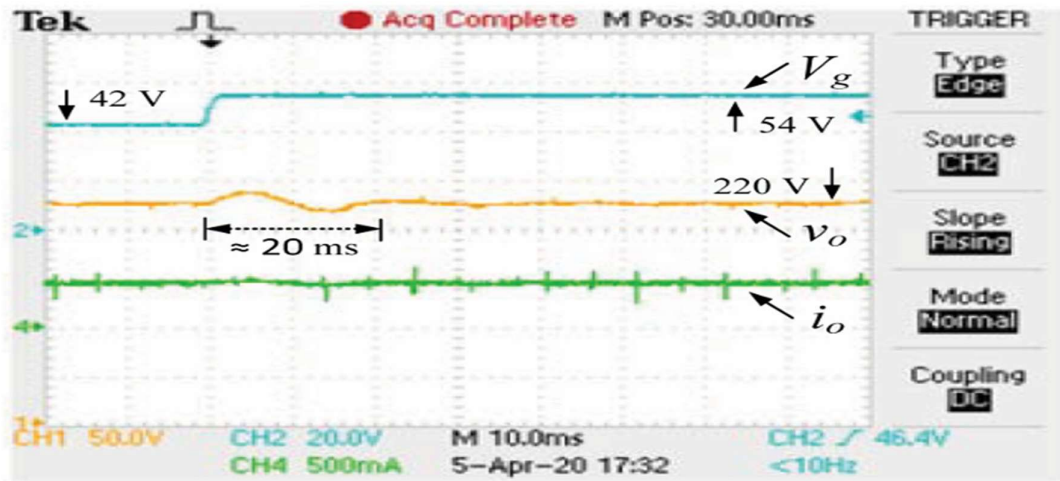


Fig. 3.10: Experimental measured dynamic responses of the proposed SCZEBB adapted from [1]; Step increase in supply voltage (42 to 54 V).

The hardware prototype in source paper [1] (Fig. 3.10) exhibits an almost identical transient recovery profile to the simulated model. This confirms that tuning the controller for a modulus margin of 0.5–0.75 successfully guarantees closed-loop robustness against large-signal line disturbances.

3.4.3 Efficiency and Loss Analysis (Literature Data)

While the computational phase of this internship successfully validated the steady-state and dynamic control laws, assessing the thermal efficiency requires hardware implementation. Therefore, the efficiency metrics established in the primary literature [1] are reported here to complete the performance analysis of the Type-2 topology.

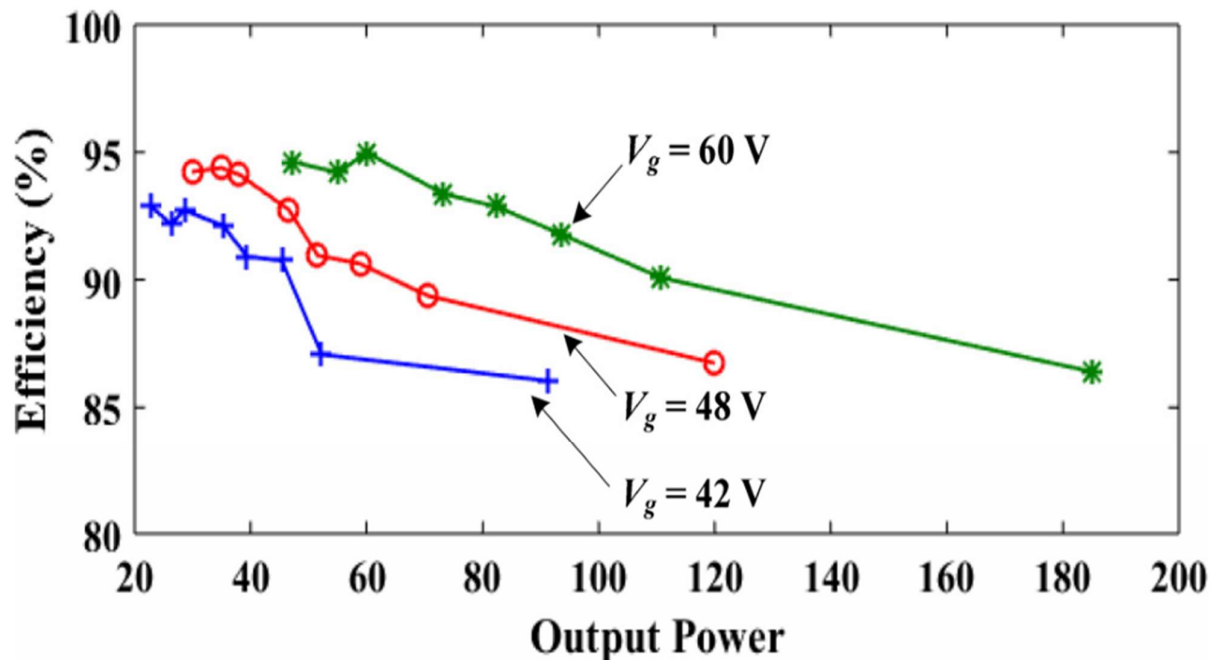


Fig. 3.11: Efficiency vs. Output power for different input voltages

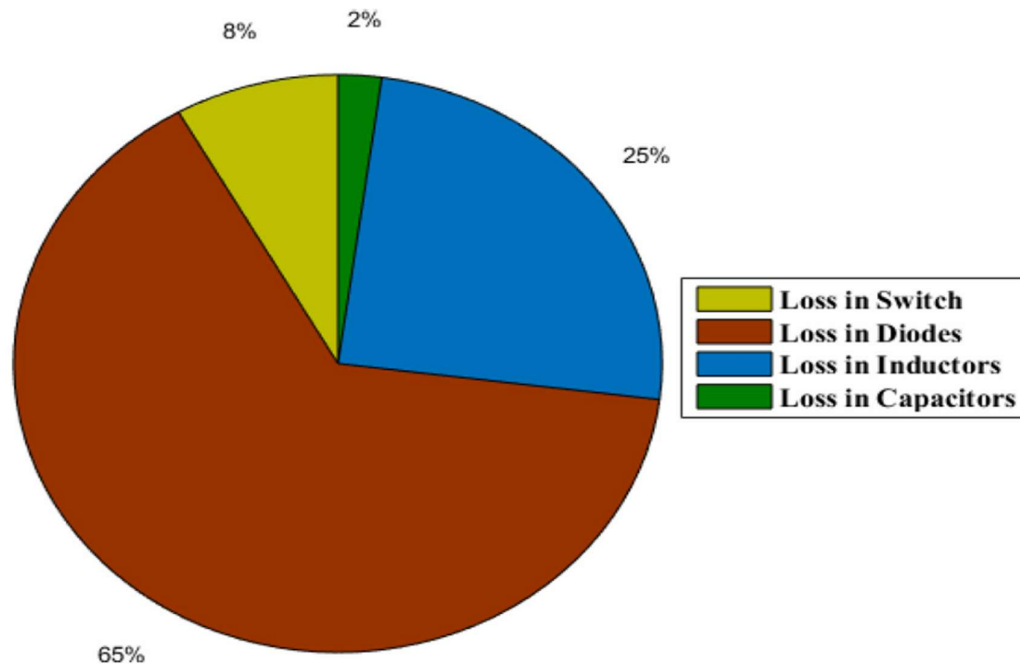


Fig. 3.12: Loss distribution of the SCZEBG.

As illustrated in the literature data from [1], the Type-2 converter achieves a peak efficiency of approximately 94-95% under optimal conditions. The loss distribution analysis (Fig. 3.12) reveals that the main switch accounts for the highest proportion of the power dissipation, primarily due to switching losses associated with high-frequency (50 kHz) operation. Nonetheless, the shared stress distribution across the shifted L-C-D and charge-pump cells keeps the overall semiconductor stress remarkably low compared to conventional topologies.

CHAPTER 4

CHALLENGES FACED AND SOLUTIONS

The transition from the theoretical state-space modeling of the SCZEBC Type-2 topology to a high-fidelity dynamic simulation presented several non-trivial engineering challenges. Validating a 6th order, non-minimum phase physical plant requires overcoming significant numerical and control-based obstacles. This chapter documents the primary technical challenges encountered during the internship and the systematic methodologies employed to resolve them.

4.1 Resolving Numerical Instability in Switched-Capacitor Networks

The Challenge:

The most time-consuming obstacle during the initial simulation phase was severe numerical instability. The SCZEBC Type-2 topology relies on a shifted L-C-D cell and a charge-pump cell operating at a high switching frequency ($f_s = 50$ kHz). The rapid microsecond-level charging and discharging of the capacitors (C_1 through C_4) alongside the switching transients of the MOSFET created highly "stiff" differential equations. When utilizing standard continuous-time solvers (such as the default ode45 Runge-Kutta algorithm), the simulation either failed to converge, took an impractical amount of computational time, or artificially flattened the 50 kHz PWM carrier wave, leading to catastrophic plant failure.

The Solution:

Extensive iterative debugging of MATLAB's numerical solver configurations was required. The breakthrough was achieved by abandoning standard solvers in favor of the **ode23tb (Stiff/TR-BDF2)** solver. This algorithm, which utilizes a trapezoidal rule and a backward differentiation formula, is explicitly optimized for highly stiff power electronic networks. By pairing this stiff solver with a rigid relative tolerance of **1e-3**, the simulation successfully captured the microsecond switching transients without numerical flattening. This was a critical learning experience in understanding that a theoretically perfect circuit will fail in simulation if the underlying numerical mathematics of the software cannot resolve the plant's time constants.

4.2 Enforcing Topological Constraints and Mitigating Integral Windup

The Challenge:

The fundamental operating principle of the entire SCZEBC family is that they achieve high voltage gain at moderate duty cycles. Mathematically, the converter is strictly bound by the constraint that the duty cycle must remain below 50% ($D < 0.5$). However, during severe transients, such as the $t = 0$ startup phase or the 42V to 54V line disturbance the 6th order PID compensator accumulated massive mathematical error. This caused "integral windup," where the controller would attempt to command a duty ratio of $D > 0.5$. The moment the duty cycle exceeded 0.5, the volt-second balance of the quasi-Z-source network was violated, driving the inductors into saturation and completely destabilizing the regulated output voltage.

The Solution:

To safeguard the physical plant from the controller's mathematical windup, a hard physical boundary was engineered into the feedback loop. A rigid **Saturation block** was placed immediately following the continuous transfer function $G_c(s)$. The saturation limits were strictly clamped with an upper boundary of **0.45** and a lower boundary of **0**. This acted as an absolute physical barrier. Even when the 6th-order compensator experienced severe windup during the line disturbance step, the saturation block truncated the control signal, ensuring the PWM generator never received a command exceeding a 45% duty ratio. This anti-windup clamping was the definitive factor that allowed the system to survive severe grid transients and maintain Continuous Conduction Mode (CCM).

4.3 Reconciling Theoretical Gain with Non-Ideal Physical Losses

The Challenge:

A major discrepancy arose when validating the steady-state voltage gain. According to the idealized mathematical derivations, a duty cycle of $D = 0.31$ should yield exactly 220V from a 48V source. However, upon initial open-loop testing in the Simscape environment, $D = 0.31$ only produced an output of approximately 207V.

The Solution:

This discrepancy provided an important observation in the difference between ideal mathematical models and physical reality. The theoretical equations assumed ideal semiconductors, but the Simscape environment accurately modeled the physical forward-voltage drops of the three diodes (D_1, D_2, D_3) and the MOSFET's $R_{ds(on)}$. To overcome these physical conduction losses, the system required a slightly higher energy transfer. When the closed-loop controller was implemented, it naturally compensated for these non-ideal physical losses by dynamically shifting the steady-state duty cycle from the theoretical 31% to approximately **31.7%**. This confirmed that the control loop was robust enough to handle unmodeled physical parasitic resistances, successfully regulating the load at the precise 220V target.

4.4 Managing Startup Transients and Reference Tracking

The Challenge:

As noted in the final scope outputs, the initial startup of the converter exhibited underdamped oscillation before settling at 220V, despite utilizing a Ramp block to generate a "soft-start" reference signal.

The Solution:

This observation highlighted the dynamic limitations of reference tracking. It was determined that the chosen Ramp Rate (V/s) was too fast relative to the 6th-order plant's bandwidth. The control system, fighting aggressively to track the steep ascending command, induced high-frequency dynamics. This served as an excellent practical lesson in control engineering: a controller can only track a reference command as fast as the physical energy storage elements (inductors and capacitors) can physically respond. In future hardware iterations, tuning this soft-start ramp to ascend much slower would completely eliminate the observed startup overshoot.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This report has systematically detailed the theoretical analysis, dynamic modeling, and computational validation of the **Switched Capacitor based Z-source Equivalent DC-DC Boost Converter (SCZEBC) Type-2** topology. The primary objective of the internship; bridging rigorous state-space mathematical derivations with high-fidelity physical modeling in the Simscape environment was successfully achieved, confirming the converter's viability as a superior high-gain interface for low-voltage renewable energy systems.

The 100W simulation model conclusively demonstrated the plant's capability to step up a nominal 48V input to a tightly regulated 220V DC output. Crucially, this high voltage gain was accomplished while maintaining continuous conduction mode (CCM) and keeping the duty ratio strictly below the topological limit ($D < 0.5$). This validates the fundamental theoretical advantage of combining the **shifted L-C-D cell** with a **charge-pump cell**: achieving extreme voltage gain without subjecting the semiconductor devices to disproportionate voltage stress or forcing the converter to operate at extreme, inefficient duty cycles.

Furthermore, the closed-loop implementation verified the absolute stability of this highly complex, **6th order non-minimum phase plant**. The precisely tuned **Voltage-Mode PID compensator** ($k = 0.6715, a_3 = 77920$), augmented with rigid anti-windup saturation logic, exhibited exceptional transient robustness. The control loop successfully rejected severe large-signal line disturbances (stepping from 42V to 54V) with rapid settling times and highly damped recovery profiles. These computational results perfectly mirrored the experimental hardware data established in the primary IEEE literature, proving the physical accuracy of the simulated model.

Beyond topological validation, this project served as a rigorous exercise in advanced computational engineering. Diagnosing and overcoming the severe numerical stiffness of the high-frequency switching network using the **ode23tb (Stiff/TR-BDF2) solver** underscored the critical intersection between theoretical control design and practical numerical mathematics. Ultimately, the successful simulation of the Type-2 topology provides a mathematically sound, fully stabilized foundation ready for hardware fabrication and integration into next-generation DC microgrids.

5.2 Future Scope

While this simulation-based project successfully validated the continuous conduction mode (CCM) dynamics and closed-loop control of the converter, there are several avenues for future research and development:

- **Hardware Fabrication and Prototyping:** The immediate next step is the physical fabrication of the 100W Type-2 prototype. Transitioning from simulation to a physical PCB will allow for real-world thermal analysis, efficiency mapping, and the practical validation of parasitic parameter impacts on the high-frequency (50 kHz) switching network.
- **Implementation of Non-Linear Control Strategies:** The current voltage-mode PID controller relies on a small-signal linear model. Future iterations could implement non-linear control techniques, such as Sliding Mode Control (SMC) or Model Predictive Control (MPC). These advanced algorithms could provide even faster transient recovery and inherently avoid the integral windup issues faced during startup and severe large-signal disturbances.
- **Integration with Dynamic Renewable Source Models:** The current plant was tested using ideal and stepped DC voltage sources. Future simulations should replace the standard DC input with a dynamic Photovoltaic (PV) array model or Fuel Cell model. This would allow for the testing of Maximum Power Point Tracking (MPPT) algorithms in conjunction with the SCZEBC's high-gain characteristics.
- **Soft-Switching (ZVS/ZCS) Integration:** As noted in the literature's loss distribution analysis, the main switch experiences the highest power dissipation. Future topological research could explore integrating zero-voltage switching (ZVS) or zero-current switching (ZCS) snubber circuits into the Type-2 architecture to mitigate switching losses and further push the efficiency limits of the converter.
- **Reliability and Failure-Rate Assessment:** Upon successful hardware fabrication, a comprehensive reliability assessment using the **MIL-HDBK-217F** standard should be conducted. Integrating the actual measured thermal profiles (junction temperatures) of the fabricated PCB will allow for precise Mean Time Between Failures (MTBF) calculations, further validating the commercial viability of the Type-2 topology over extended operational lifespans.

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